

Day 8 : Navigation And Routing

1. Introduction to Navigation

Navigation is the process of moving from one screen (page) to another screen in a Flutter application. Almost every mobile application uses navigation. For example:

- Login → Home Screen
- Home → Profile Screen
- Home → Settings Screen
- Product List → Product Details

Flutter provides a navigation system through the **Navigator** class, which manages a stack of screens.

2. Why Do We Need Navigation?

Without navigation, all content would have to be displayed on a single screen, making the application difficult to use.

Benefits:

- Better user experience
- Organized application structure
- Easy movement between screens
- Supports large applications
- Allows data sharing between screens

3. Navigator

The Navigator widget manages a stack of routes (screens).

Think of it like a stack of books:

- New screen opened → added on top
- Back button pressed → top screen removed

Profile Screen

Home Screen

Login Screen

4. Navigator.push()

Used to open a new screen.

```
Navigator.push(  
  context,  
  MaterialPageRoute(  
    builder: (context) => SecondScreen(),  
  ),  
);
```

Working

1. User clicks a button.
2. Flutter creates a new route.
3. Route is added to Navigator stack.
4. New screen appears.

```
ElevatedButton(  
  onPressed: () {  
    Navigator.push(  
      context,  
      MaterialPageRoute(  
        builder: (context) => SecondScreen(),  
      ),  
    );  
  },  
  child: Text("Open Second Screen"),  
)
```

5. Navigator.pop()

Used to close the current screen and return to the previous screen.

7. createState() Method

```
ElevatedButton(  
  onPressed: () {  
    Navigator.pop(context);  
  },  
  child: Text("Back"),  
)
```

6. MaterialPageRoute

A route that provides Material Design page transition animation.

```
MaterialPageRoute(  
  builder: (context) => HomeScreen(),  
)
```

Features

- Automatic transition animation
- Platform support
- Easy implementation
- Material Design compliant

7. Passing Data Forward

Often one screen needs information from another screen.

```
Navigator.push(  
  context,  
  MaterialPageRoute(  
    builder: (context) =>  
      SecondScreen(name: "Sneha"),  
  ),  
);
```

```
class SecondScreen extends StatelessWidget {  
  final String name;  
  
  SecondScreen({required this.name});  
  
  @override  
  Widget build(BuildContext context) {  
    return Scaffold(  
      body: Center(  
        child: Text(name),  
      ),  
    );  
  }  
}
```

8 . Returning Data Backward

Sometimes a screen sends information back.

Return Data :

```
Navigator.pop(  
  context,  
  "Mumbai",  
);
```

Receive Data :

```
final result = await Navigator.push(  
  context,  
  MaterialPageRoute(  
    builder: (context) => CityScreen(),  
  ),  
);  
  
print(result);
```

9. Named Routes

Instead of creating routes repeatedly, Flutter allows route names

Define Routes :

```
MaterialApp(  
  routes: {  
    '/home': (context) => HomeScreen(),  
    '/profile': (context) => ProfileScreen(),  
    '/settings': (context) => SettingsScreen(),  
  },  
);
```

Navigate :

```
Navigator.pushNamed(  
  context,  
  '/profile',  
);
```

10. Passing Arguments in Named Routes

Sending Data :

```
Navigator.pushNamed(  
  context,  
  '/profile',  
  arguments: "Sneha",  
);
```

Receiving Data :

```
final name =  
ModalRoute.of(context)!  
.settings.arguments as String;
```

11. Navigation Drawer

A side menu that slides from the left side of the screen.

Used in apps like:

- Gmail
- Facebook
- Amazon

Drawer Structure :

```
Drawer(  
  child: ListView(  
    children: [  
      DrawerHeader(  
        child: Text("User"),  
      ),  
      ListTile(  
        title: Text("Home"),  
      ),  
      ListTile(  
        title: Text("Profile"),  
      ),  
    ],  
  ),  
)
```

Drawer Header :

```
DrawerHeader(  
  child: Text("Welcome User"),  
)
```

Handling Drawer Taps :

```
ListTile(  
  title: Text("Profile"),  
  onTap: () {  
    Navigator.pushNamed(  
      context,  
      '/profile',  
    );  
  },  
)
```

12. Bottom Navigation Bar

A navigation bar placed at the bottom of the screen.

Common Apps:

- Instagram
- WhatsApp
- YouTube

Components :

1. Current Index

```
int selectedIndex = 0;
```

2. Navigation Items :

```
BottomNavigationBarItem(  
  icon: Icon(Icons.home),  
  label: "Home",  
)
```

Example :

```
BottomNavigationBar(  
  currentIndex: selectedIndex,  
  onTap: (index) {  
    setState(() {  
      selectedIndex = index;  
    });  
  },  
  items: [  
    BottomNavigationBarItem(  
      icon: Icon(Icons.home),  
      label: "Home",  
    ),  
    BottomNavigationBarItem(  
      icon: Icon(Icons.person),  
      label: "Profile",  
    ),  
  ],  
)
```

Code :

